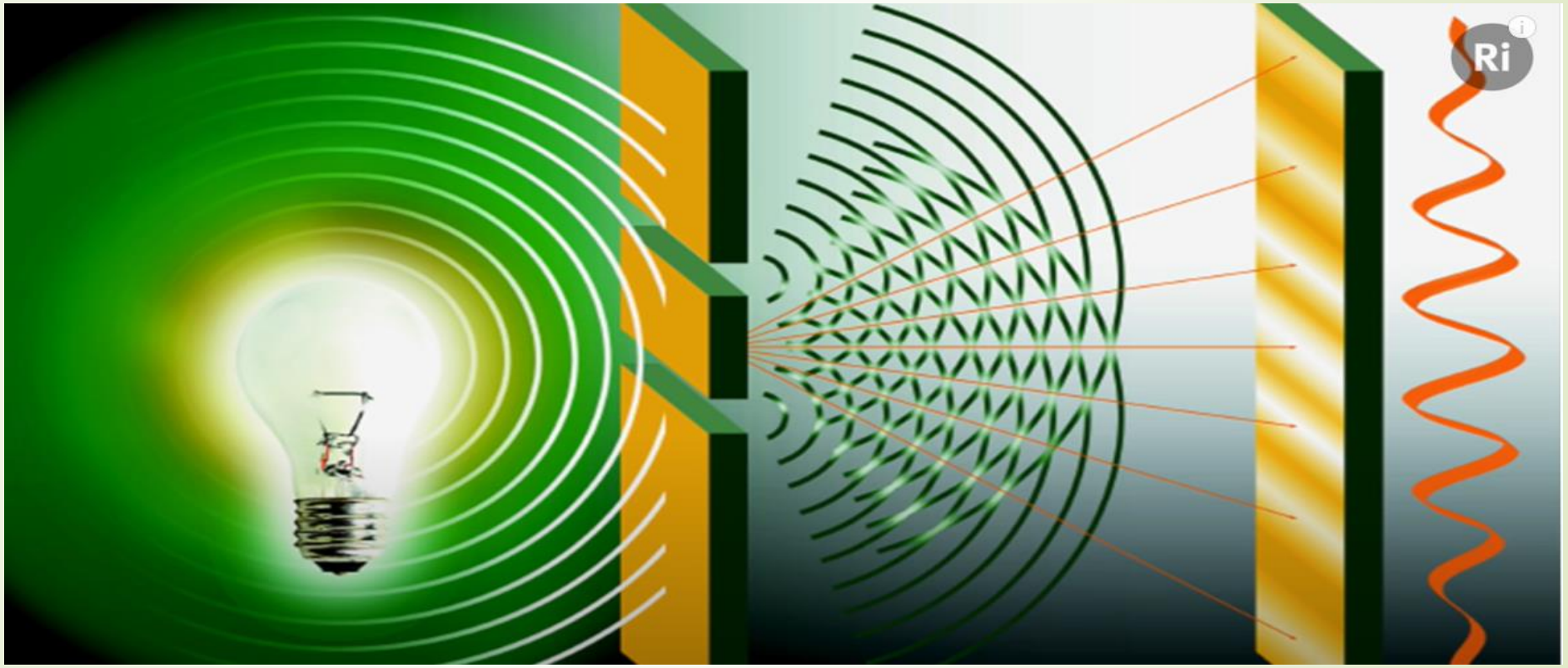
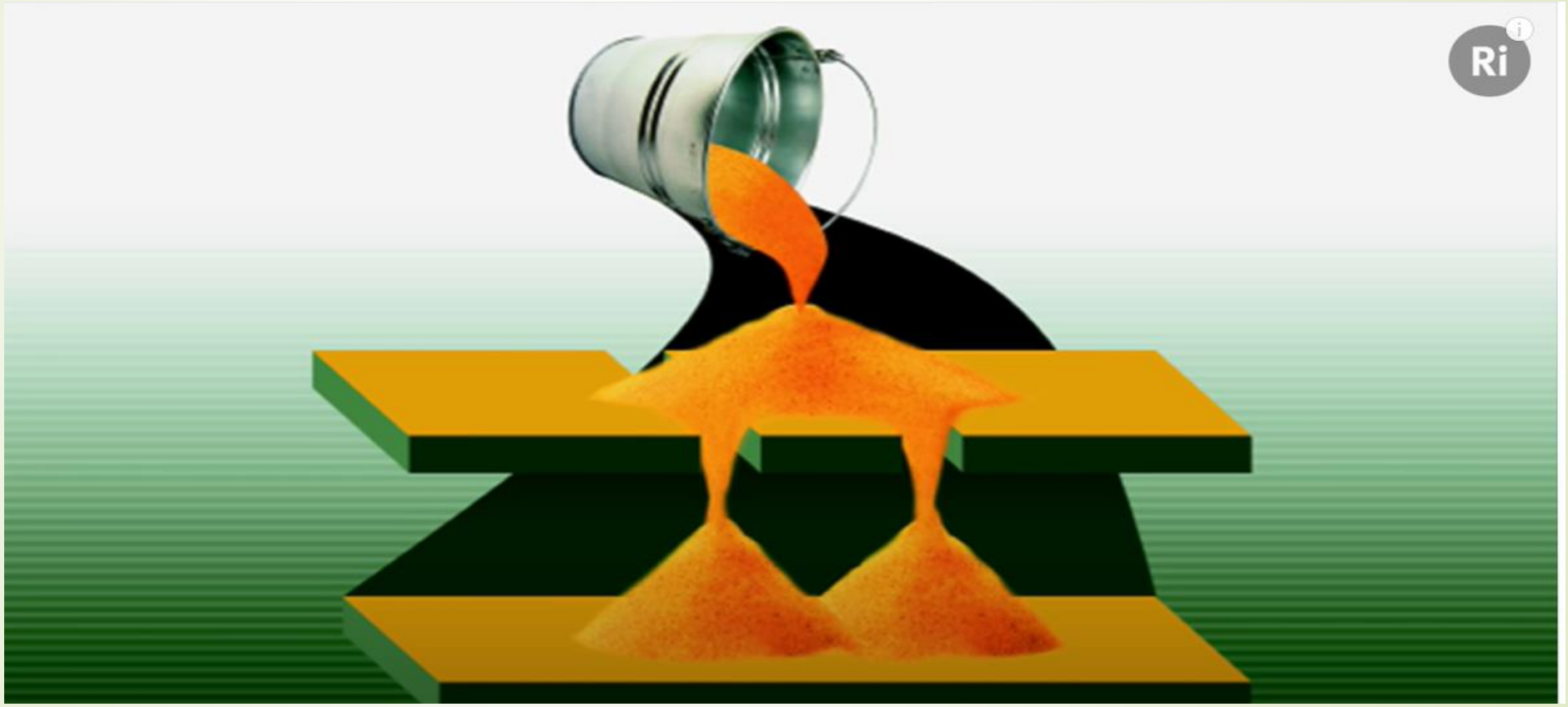
Quantum Physics

Dr. Sreemoyee Sarkar

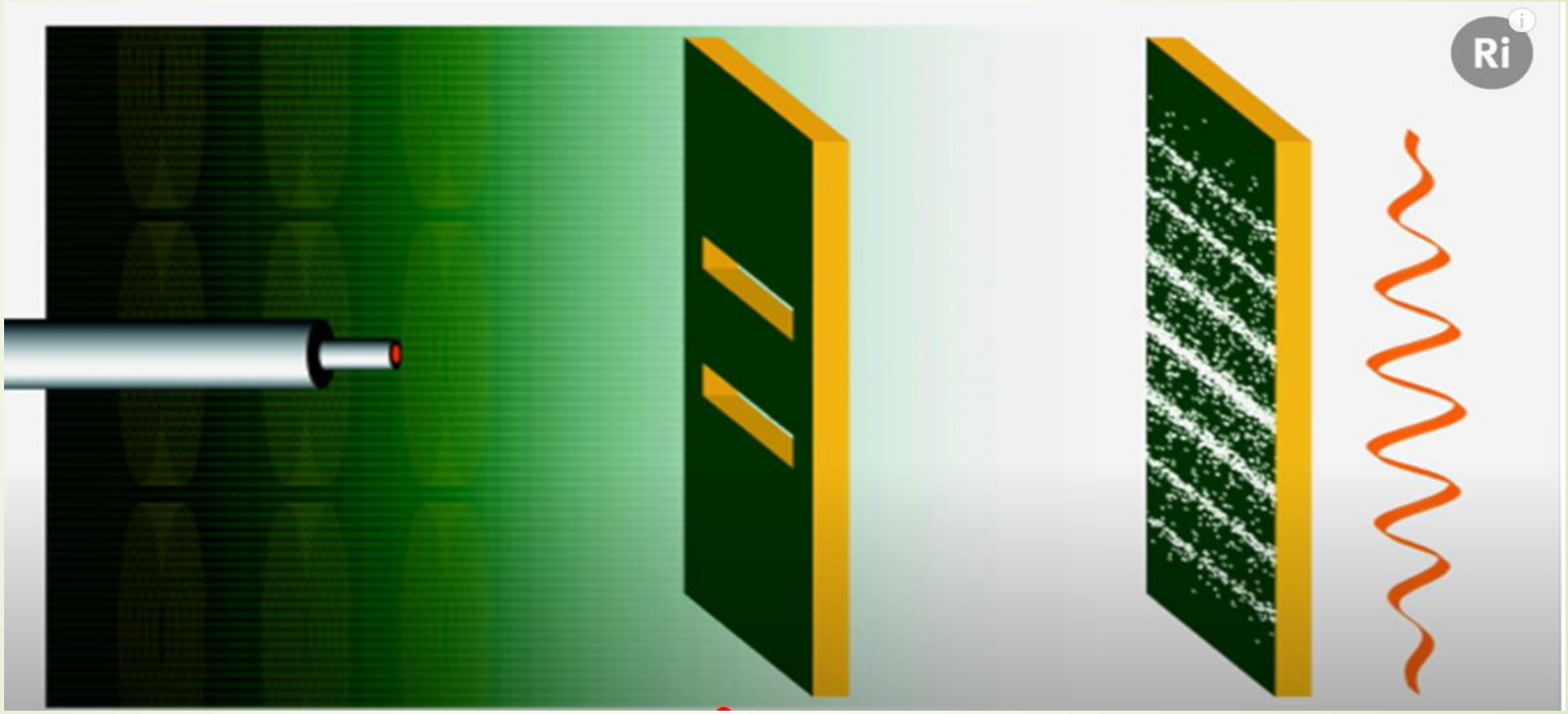
Double Slit Interference of light wave



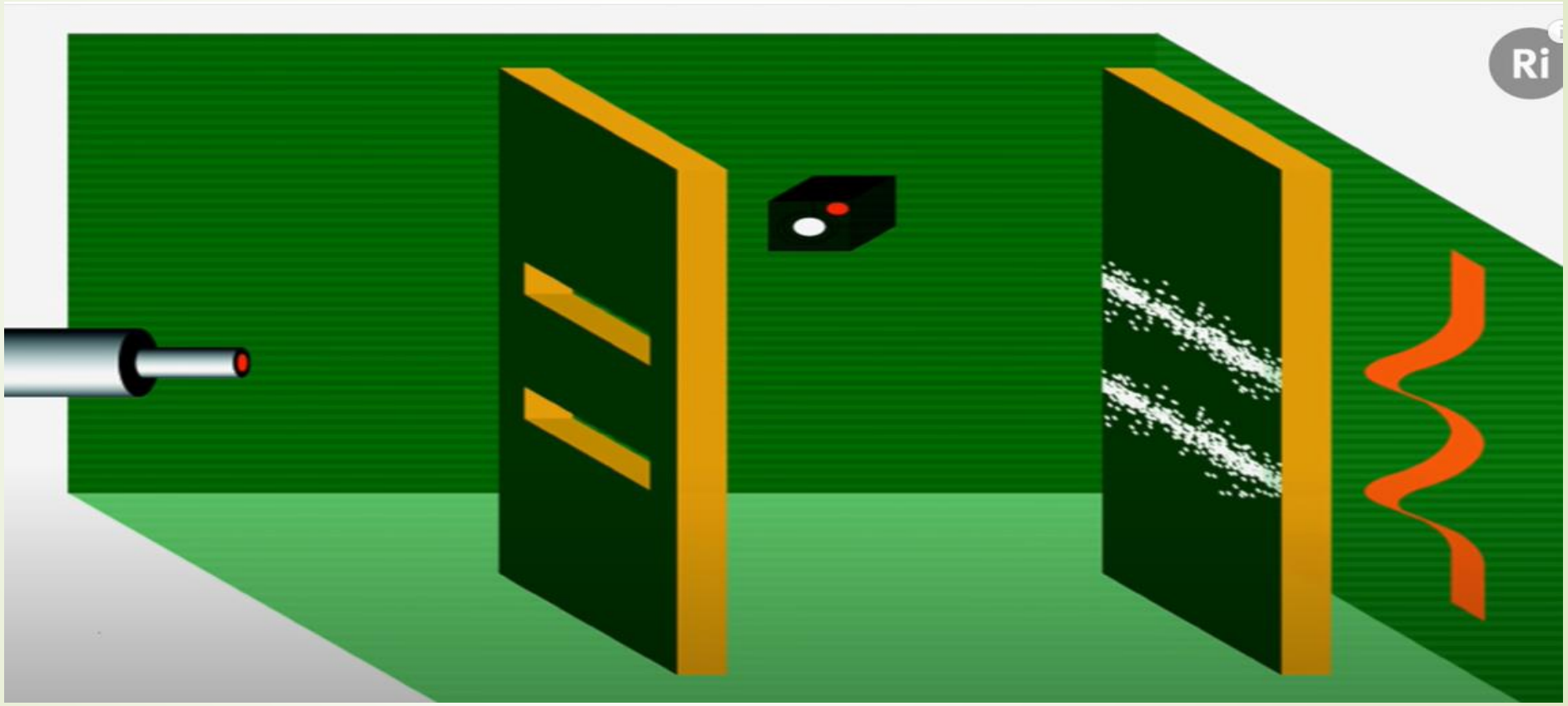
Double slit experiment with sand



Double slit interference experiment of electrons



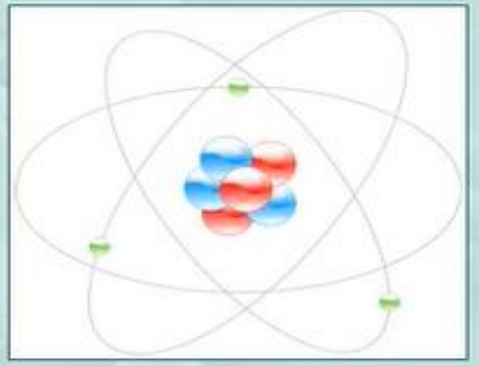
Double slit interference experiment of electrons



Introduction

The foundations for many of the technological advances in our world today were laid by the scientists Max Planck and Albert Einstein early in the 20th century.

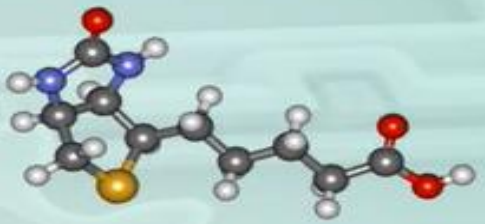
Their studies led to the recognition that energy is emitted and absorbed by atoms in discrete amounts – not in a linearly variable manner. This discovery of discrete energy quanta led to the study of quantum physics.



Progress in today's world, full of microelectronic devices, is driven by this branch of science.

Electronics

Much of modern technology operates at a scale where quantum effects are significant.



The study of semiconductors led to the invention of the diode, the transistor, and the microprocessor, which are indispensable for modern electronics.



Other examples include the laser, the electron microscope, and magnetic resonance imaging.



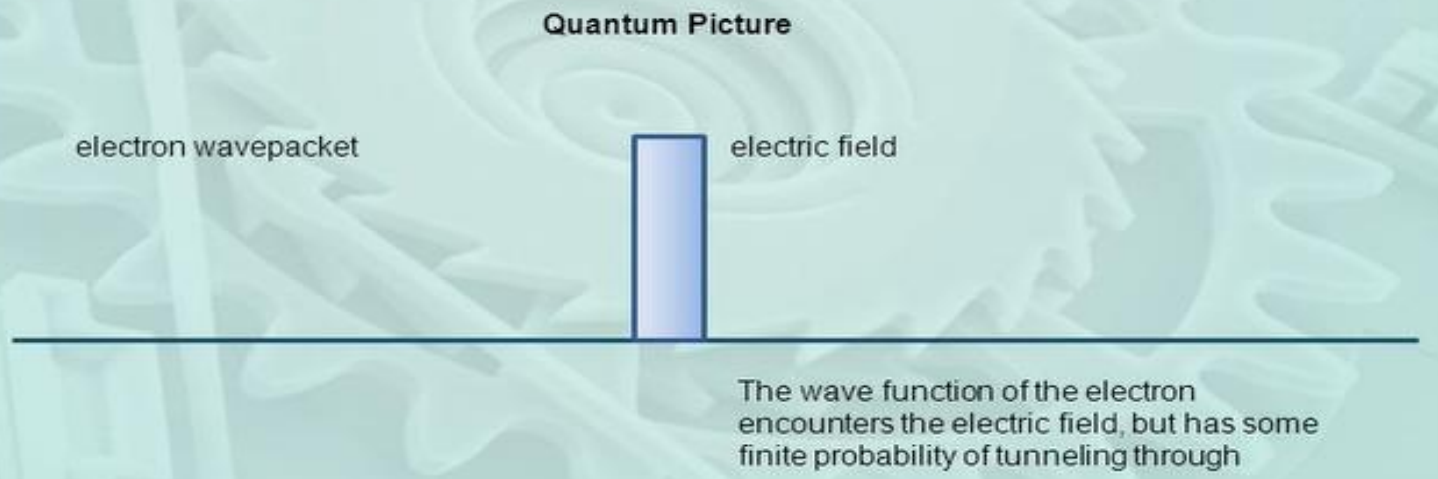
SlidePlayer

Next >

Quantum Tunneling

Quantum tunnelling is vital in many electronic devices, being the basis for transistor operation.

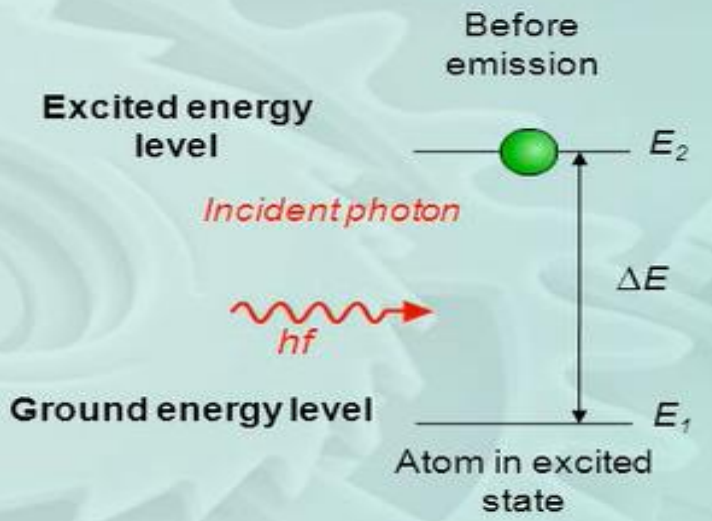
Flash memory chips, found in USB drives, use quantum tunnelling to erase their memory cells.



Lasers

Lasers work using the quantum phenomenon known as stimulated emission.

The electron of an atom is stimulated by an incident photon.

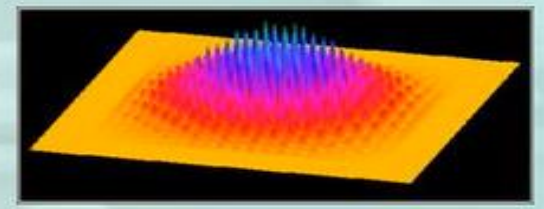


$E_2 - E_1 = \Delta E = hf$

Quantum Dots

SlidePlayer

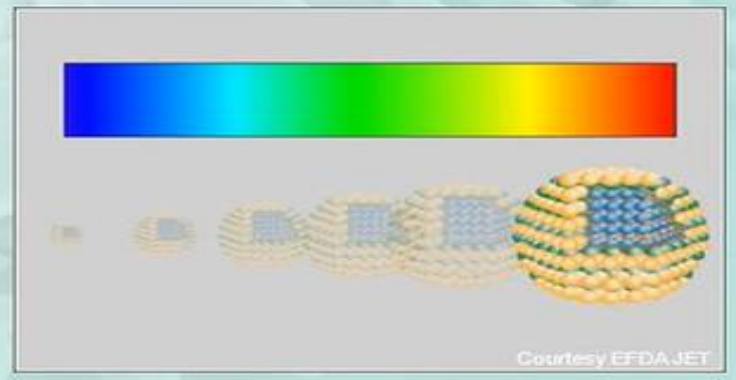
Quantum dots are tiny particles of a semiconductor material, with a width of about 50 atoms.



By controlling the size of the dot, the light it emits or absorbs can be very precisely controlled.

Applications of quantum dots include the following:

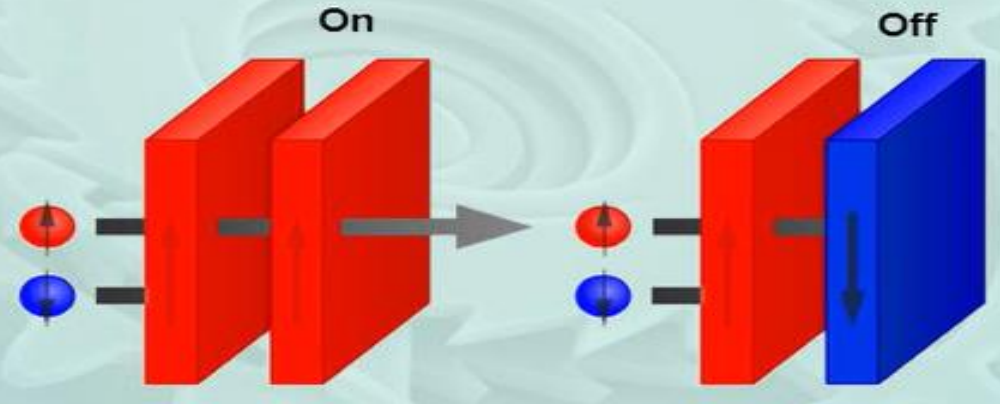
- Lighting
- Solar cells
- Light detectors
- Security marking



Next >

Spin

Electron spin plays an important role in magnetism, with applications in computer memories. The manipulation of nuclear spin by radio frequency waves is important in medical imaging.



SlidePlayer

Next >

Introduction to Quantum Mechanics

Quantum mechanics is one of the most important intellectual achievement of the 20th century. This sophisticated field in physics has refined our understanding of the nature, particularly in the nanometer length scale down to atomic and subatomic level and has profound implications in chemistry, materials science, optics, and electronics. The existence of orbitals and energy levels in atoms can only be explained using quantum mechanics. It can aptly explain the behaviors of insulators, semi-conductors and conductors, as well as that of giant magneto-resistance. It can explain the quantization of light and both its corpuscular and wave nature.

Quantum mechanics has found myriad of applications in photonics, quantum electronics, and micro-electronics, nanotechnology and many more emerging technologies. Therefore, it is of extreme importance that scientists and engineers learn quantum mechanics.

Historical background

A series of experiments during the late nineteenth century, with puzzling and paradoxical results, proved impossible to be explained by the classical physics of mechanics and electromagnetism; and needed radically new concepts of theoretical physics. These experiments fell roughly into three categories:

1. *The Corpuscular (particle-like) Behaviour of Light Waves*
   Black-body radiation, the photoelectric effect, the Compton effect.
2. *The Stability of the Atom*
   Why doesn't the electron fall into the nucleus as predicted by classical electromagnetic theory? What is the origin of atomic spectra?
3. *The Wave-like Behaviour of Particles*
   Electron diffraction

The emergence of quantum mechanics was an attempt to explain these phenomena and it has proved itself to be immensely accurate.

In the following sections we shall discuss these with considerable details.

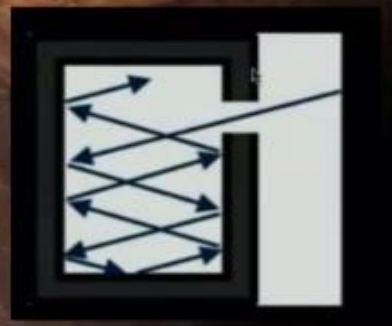
Module 1: Quantum Physics

- Motion of macroscopic particles can be explained by classical theory of Mechanics.
- But it fails to explain the motion of microscopic particles like electron, proton etc.
- **Quantum mechanics** was developed from Quantum theory to explain the properties associated with such particles.

Ref: Online source

DEFINATION OF BLACKBODY

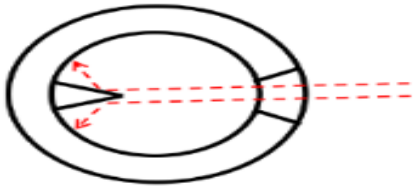
A black body is an ideal body which allows the whole of the incident radiation to pass into itself (without reflecting the energy) and absorbs within itself this whole incident radiation (without passing on the energy). Is called a Blackbody.



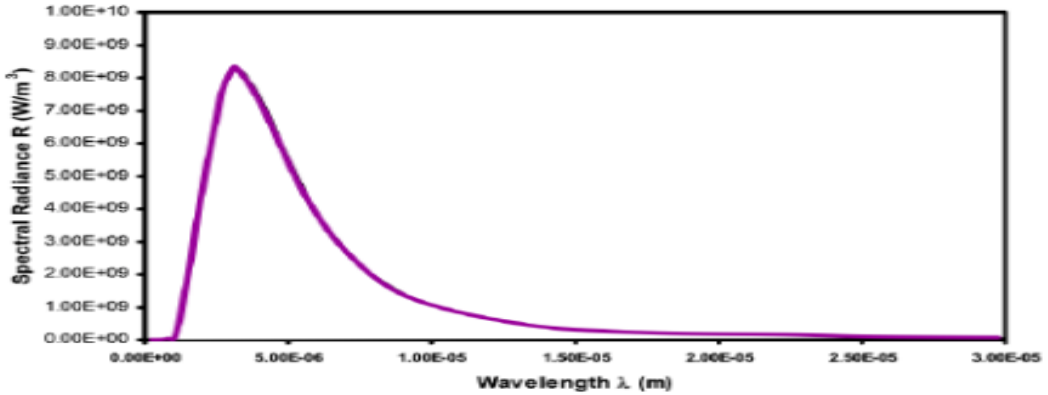
Ref: Online source

Black-body radiation

All bodies give out radiation consistent with the temperature it is at. For example, at room temperature, humans give out infra-red (IR) radiation. This is the reason that militaries use IR goggles to spot people at night. At around 1000°C, bodies give out visible light, which is how conventional light bulbs function. When electromagnetic radiation is incident on a substance, some of it will be absorbed, some reflected and some transmitted. A substance can be imagined and constructed that absorbs all radiation incident on it as long as it cooler than its surroundings. This substance will also emit radiation as long as it is hotter than its surroundings. Such a substance is referred to as a "Black body". Graphite, as a material, comes close to satisfying this description. The figure below shows the schematic of the blackbody designed by Kirchoff. The arrows indicate that radiation entering the body will get absorbed by the internal surfaces of the body. Kirchhoff's law, states that the emissivity of a body which is in thermal equilibrium with its surrounding is equal to the absorptivity of the body.

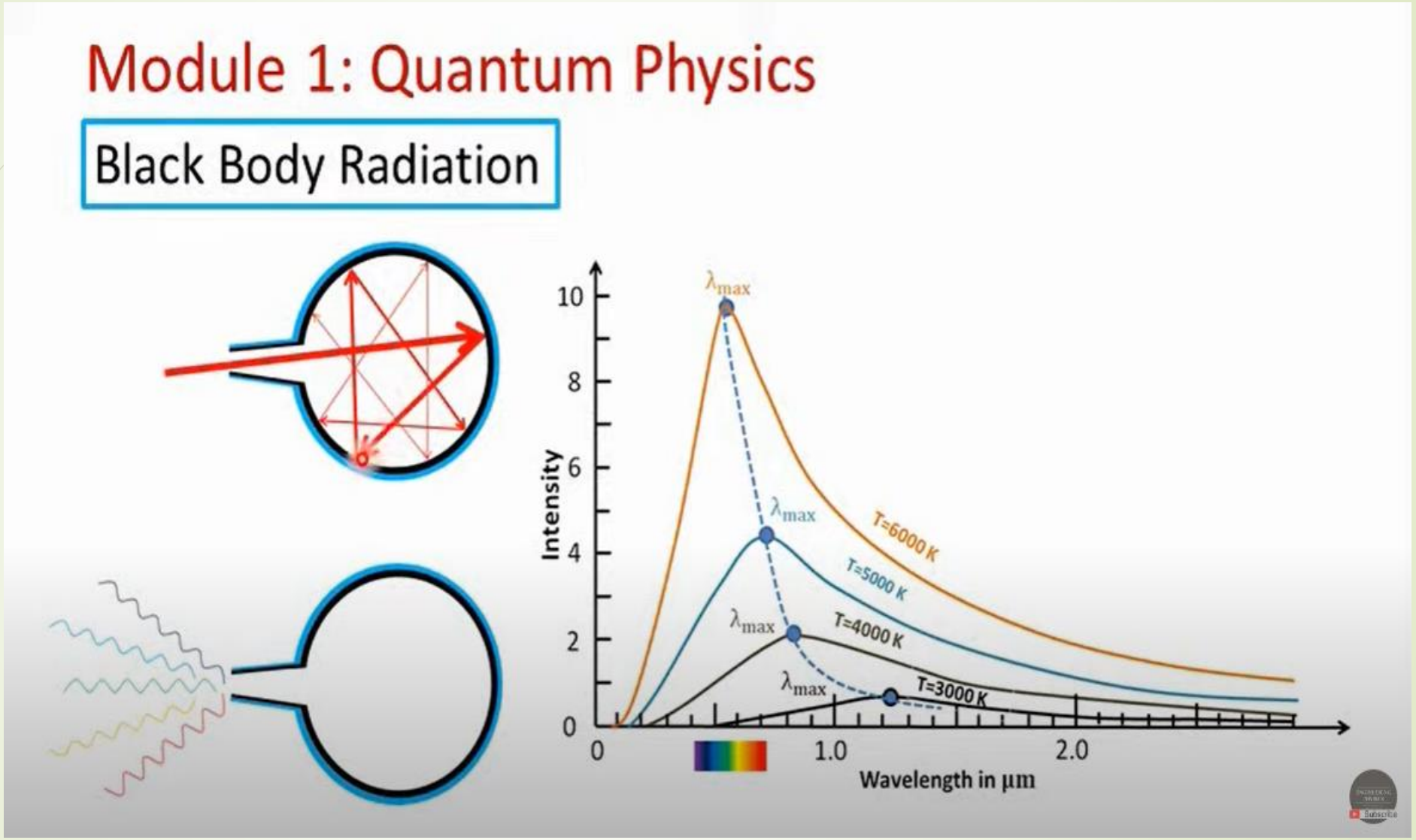


In general, electromagnetic radiation emitted by a blackbody comes out over a range of wavelengths, however it is not emitted with uniform intensity across all wavelengths. The maximum intensity of the radiation occurs at one wavelength and the intensity decreases for all other wavelengths. The spectral distribution of the radiation emitted by blackbody is shown in figure below.



Module 1: Quantum Physics

Black Body Radiation



Ref: Online source

The spectral radiance plotted on the y-axis represents power per unit area per unit wavelength and is therefore represented by $W/m^2/m = W/m^3$, which is the unit shown on the y-axis. The area under the curve represents intensity (I), which is power per unit area.

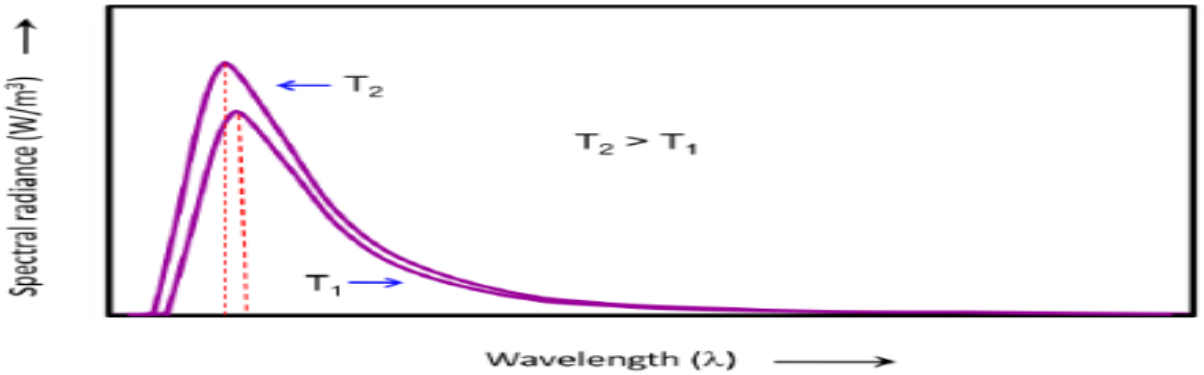
Mathematically, $I = \int_0^\infty R(\lambda)d\lambda$

There are two observations that can be made about blackbody radiation:

1) As temperature T of the body increases, intensity of the radiation from the body increases.

2) Higher the temperature, lower is the wavelength of the most intense part of the spectrum.

These two observations are represented by the figure below:



From this figure, it is evident that at the higher temperature T_2, the area under the curve, and hence intensity, has increased relative to the curve at T_1. Also, at the higher temperature T_2, the wavelength corresponding to the maximum intensity (identified using the red dotted lines in the figure), has decreased relative to that at T_1.

These two trends in blackbody radiation, can be mathematically stated in the form of two laws:

Stefan Boltzmann Law

Josef Stefan based on experimental facts suggested that the total emissive power of a blackbody (I) is proportional to the fourth power of the absolute temperature.

$$I(T) = \sigma T^4$$

Here, I is the total amount of radiation emitted by an object per square meter (Wm^{-2}), σ is a constant called the Stefan-Boltzmann constant (5.67 x 10-8 $Wm^{-2}K^{-4}$), and T is the absolute temperature of the body (in K).

Wein's displacement law

The variation of blackbody radiation with temperature shows that as the temperature increases the most intense part of the spectrum shifts towards the shorter wavelength. This is given by Wein's law

$$\lambda_{max} T = constant = 2898\ \mu mK$$

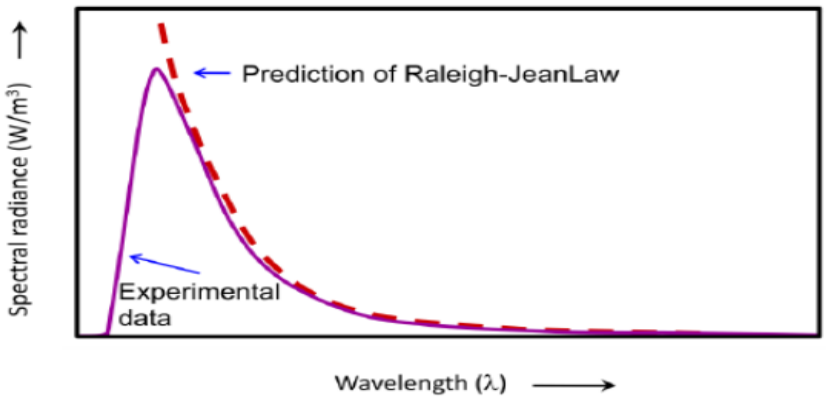
The scientific challenge that remained was to determine the exact form of the spectral radiance, or power per unit area at a particular wavelength, R (λ).

Several researchers worked to determine the form of R (λ). One of the early attempts, looked at the matter-radiation interaction in a classical manner, i.e. assumed an equipartition of energy, wherein all modes available to the solid through which it could absorb energy, participated in the process equally. This led to the law known as the Rayleigh-Jean law, which provides an equation for the spectral radiance as follows:

$$R(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

$$R(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

At higher values of λ, this led to a good match between theory and experiment. However as λ decreases, the theory predicts an ever increasing spectral radiance – a prospect dubbed as "Ultraviolet catastrophe". Common experience shows that this does not occur – bodies do not spontaneously release infinite energy. Therefore, the Rayleigh-Jean law comprehensively fails at lower wavelengths. The mismatch between theory and experiment is shown in the schematic in figure below.



Hence, classical wave theory of radiation is not able to predict blackbody radiation. In order to explain these facts Max Planck (1901) gave a theory called Planck's quantum theory of radiation.

Equations of Planck's law

We have two forms, As a function of frequency ,

$$\rho_{(v,\ T)}\mathrm{d}v = \frac{8\pi h v^3}{C^3}\ \frac{dv}{\frac{hv}{e^{kT}} - 1}$$

As a function of wavelength,

$$\rho_{(\lambda,\ T)}\mathrm{d}\lambda = \frac{8\pi h v^3}{\lambda^5}\ \frac{d\lambda}{\frac{hv}{e^{kT}} - 1}$$

Where ρ (v T)dv = radiant energy density

c is the speed of light,

h = Planck constant,

k =Boltzmann constant,

v = frequency ($c = \lambda v$)

λ = Wavelength

Ref: Online source

The various postulates of quantum theory of radiations are as follows:

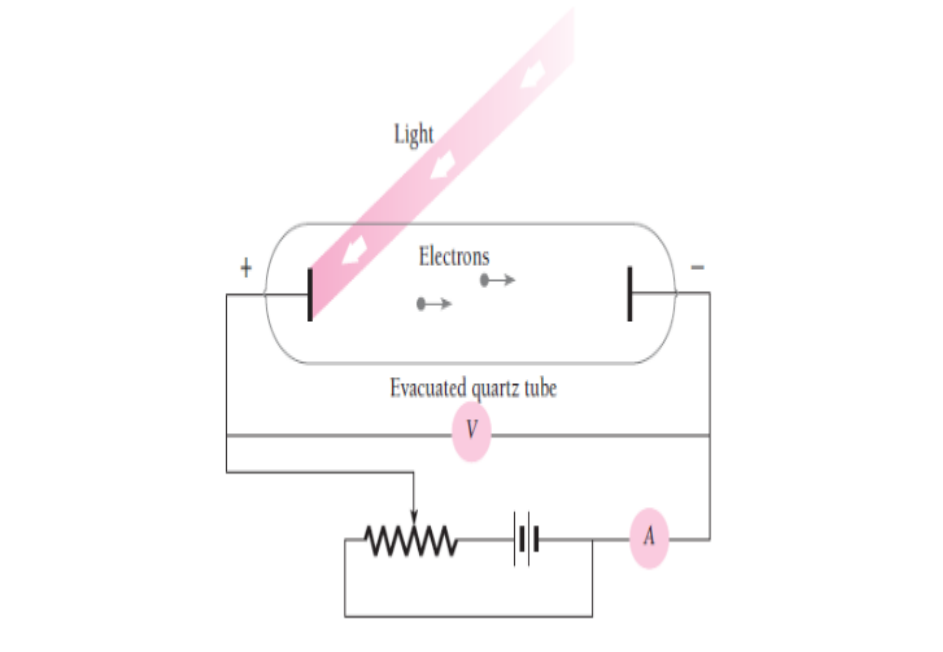
A radiation has energy. As light and heat are radiations, they are also associated with energy.

Radiant energy is not emitted or absorbed continuously but discontinuously in the form of small packets called photons. Photon is not a material body but is considered to be a massless packet of energy.

The energy E of a photon is related to the frequency of radiation, ν, by the equation $E = h\nu$, where h is Planck's constant.

Whenever a body emits or absorbs energy, it does so in whole number multiples of photons, i.e., $\boldsymbol{E = nh\nu}$, where $n = 1, 2, 3$ etc.

Photoelectric Effect



- The photoelectric effect was first observed by Heinrich Hertz in 1887. When polished metal plates are irradiated, he observed, they may emit electrons, then called "photo-electrons". The emitted electrons thus produce a photoelectric current.
- If we illuminate one of the metal plates with photons of energy above the work function value of the metal, electrons absorb the photons and escape from the surface of the metal.
- As the plate is within an evacuated tube these electrons will move off the surface with whatever kinetic energy they possess, in whatever direction they are emitted.
- Some will move across the tube and hit the metal plate at the opposite end of the tube and the ammeter will register a current. The current gives no indication of the kinetic energy of the photoelectrons, only the no. of them released per second.

- The energy they possess is kinetic energy
- We apply an electric voltage across the tube to oppose the travel of the electrons between electrodes. Initially on application of a low negative potential, electrons are repelled and some of them go back. But others having high kinetic energy are still able to reach the plate. As the negative potential increases, less and less electrons are able to reach the other side of the tube and finally when no one can reach, the photocurrent becomes zero. This pot. difference is called as stopping potential and is related to the maximum kinetic energy of the ejected photoelectron. The fastest electron will be able to reach the negative plate just before the photocurrent reaches zero.

1. Within the limits of experimental accuracy, there is no time interval between the arrival of light at a metal surface and the emission of photoelectrons
2. A bright light yields more photoelectrons than a dim one of the same frequency, but the electron energies remain the same. The em theory of light, on the contrary, predicts that the more intense the light, the greater the energies of the electrons.
3. The higher the frequency of the light, the more energy the photoelectrons have. Blue light results in faster electrons than red light.
4. At frequencies below a certain critical frequency 0, which is characteristic of each particular metal, no electrons are emitted. Above 0 the photoelectrons range in energy from 0 to a maximum value that increases linearly with increasing frequency. This observation cannot be explained by the em theory of light.

The experimental results can be summarized as

1. *It is an instantaneous process*. The time lag between the incidence of light and emission of electrons is ~ 10^{-9} s.
2. *There is a threshold frequency for the emission of photoelectrons*. If the light beam contains no photons having frequencies higher than this threshold, no photoelectron will be emitted (in other words, there will be no photo current) *regardless of the intensity of the light source*.
3. *The threshold frequency varies with different metals*.

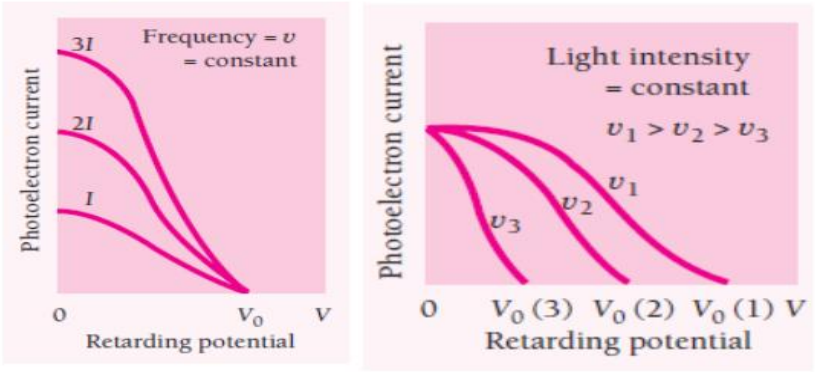
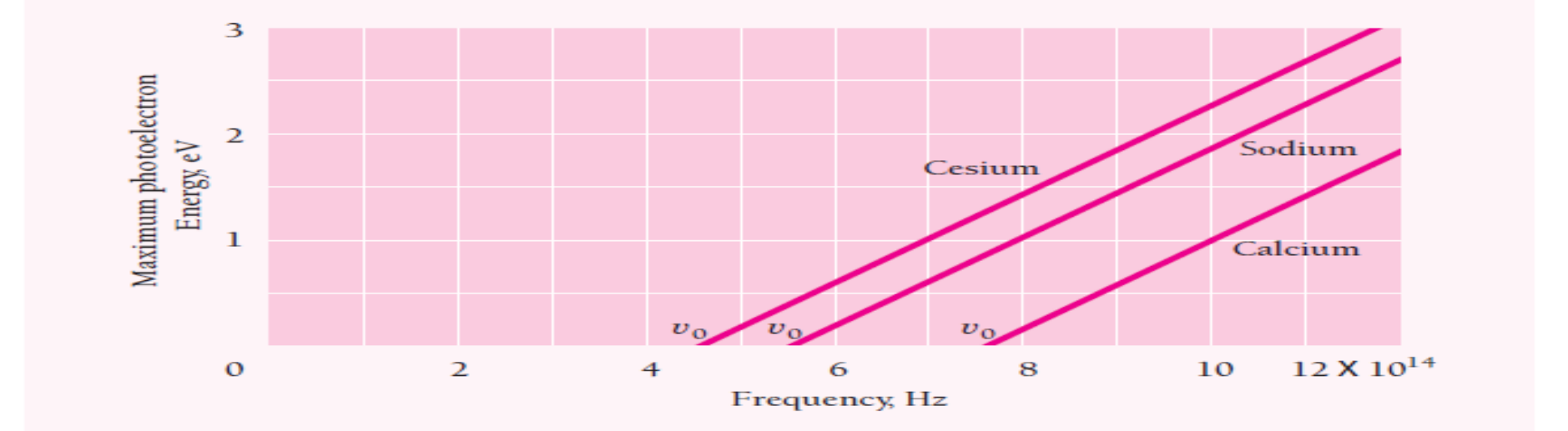


Fig. 2. Retarding potential vs. photocurrent graph at (a) constant frequency and (b) constant light intensity.

4. *The magnitude of the photocurrent is independent of the frequency of the light but is directly proportional to the intensity of the light. However, the average electron energy remains same* (Fig. 2a).
5. *For a given material, the higher the frequency of the light, the more energy the photoelectrons have and the value of stopping potential also increases* (Fig. 2b).

Quantum Theory of Light

- In 1905, Einstein realized that the photoelectric effect could be understood if the energy in light is not spread out over wavefronts but is concentrated in small packets, or **photons.** (The term photon was coined by the chemist Gilbert Lewis in 1926.) Each photon of light of frequency has the energy $h\nu$, the same as Planck's quantum energy.
- Em wave energy is concentrated in photons and not spread out, there should be no delay in the emission of photoelectrons.
- All photons of frequency have the same energy, so changing the intensity of a monochromatic light beam will change the number of photoelectrons but not their energies.
- The higher the frequency , the greater the photon energy $h\nu$ and so the more energy the photoelectrons have.



There must be a minimum energy for an electron to escape from a particular metal surface. This energy is called the **work function** of the metal.

The greater the work function of a metal, the more energy is needed for an electron to leave its surface, and the higher the critical frequency for photoelectric emission to occur.

$$W = h\nu_0$$

According to Einstein, the photoelectric effect in a given metal should obey the equation $h\nu = K.E_{max} + W$

KEmax is the maximum photoelectron energy (which is proportional to the stopping potential)

The Compton Effect

- Compton carried out experiments (1923–1924) scattering X-rays off a carbon target. X-rays correspond to photon energies in the range from 100 eV to 100 KeV. The goal was scattering X-ray photons off free electrons, and with some qualification, the electrons in the atoms behave this way.
- According to the classical electromagnetism, the charged electron will begin to oscillate at the frequency of the incident wave, and will therefore radiate further electromagnetic waves at exactly the same frequency as the incident wave in all directions.

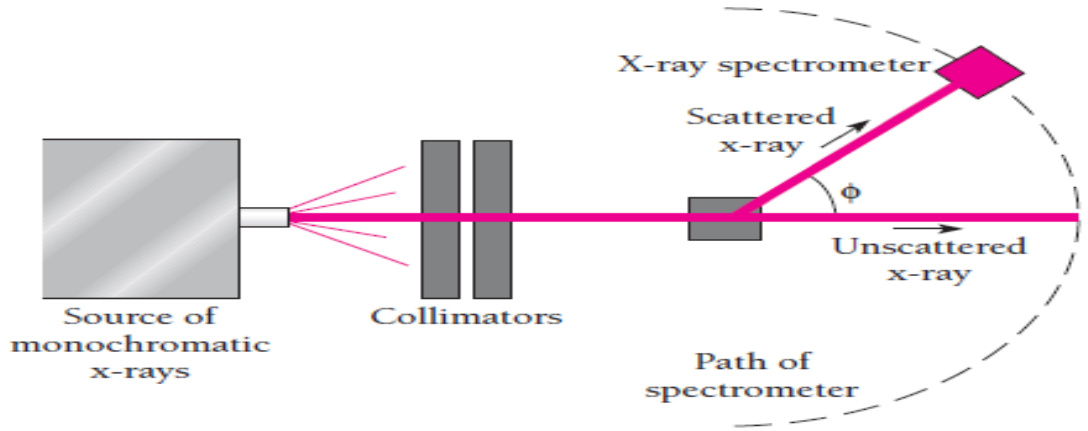


Figure 2.23 Experimental demonstration of the Compton effect.

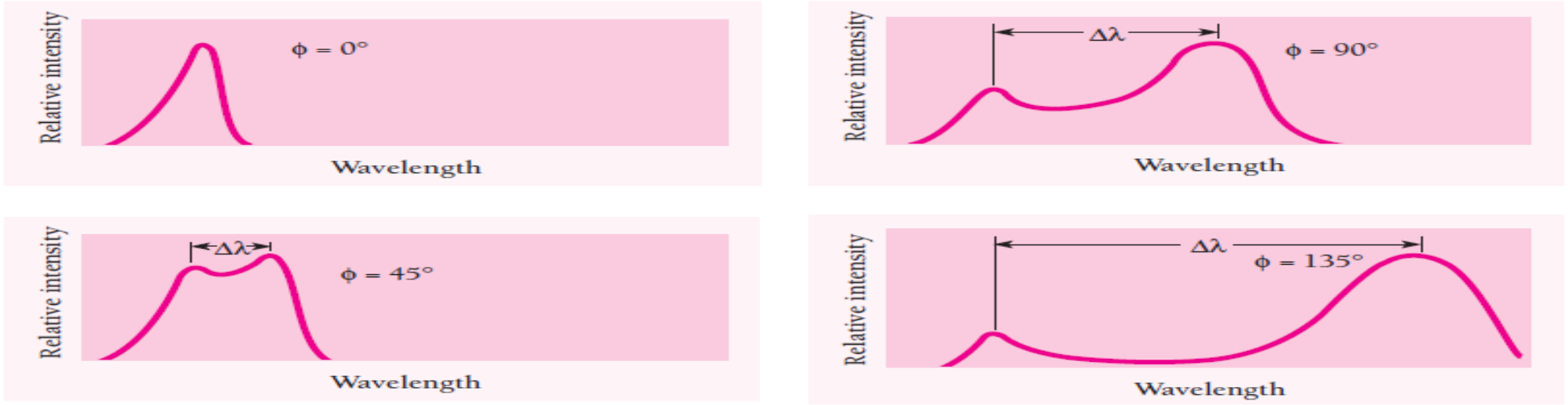


Figure 2.24 Experimental confirmation of Compton scattering. The greater the scattering angle, the greater the wavelength change, in accord with Eq. (2.21).

Loss in photon energy = gain in electron energy

$$h\nu - h\nu' = \text{KE} \tag{2.14}$$

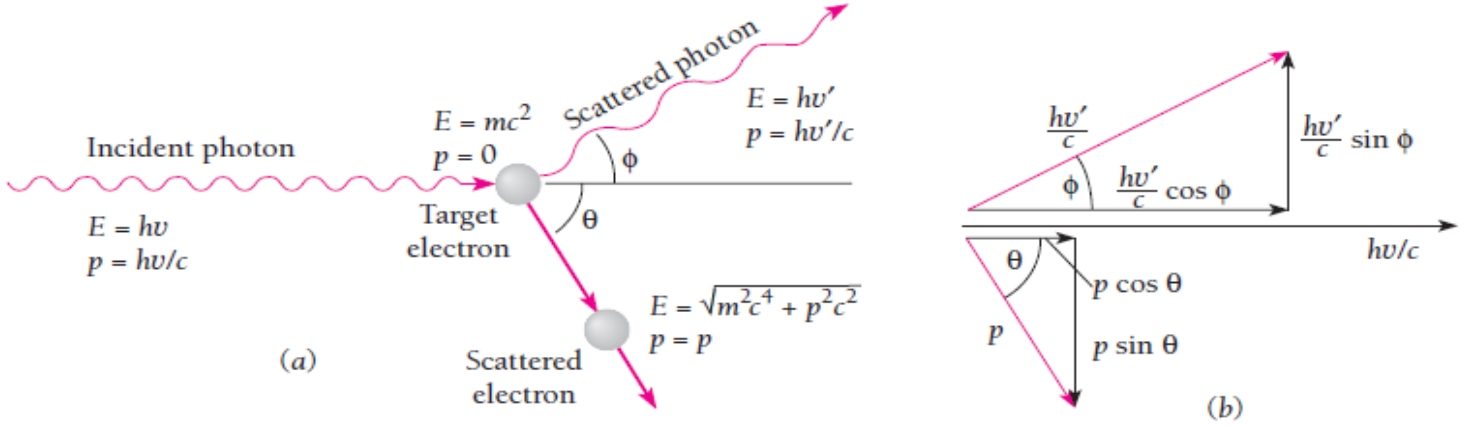
From Chap. 1 we recall that the momentum of a massless particle is related to its energy by the formula

$$E = pc \tag{1.25}$$

Since the energy of a photon is $h\nu$, its momentum is

Photon momentum

$$p = \frac{E}{c} = \frac{h\nu}{c} \tag{2.15}$$



The initial photon momentum is $h\nu/c$, the scattered photon momentum is $h\nu'/c$, and the initial and final electron momenta are respectively 0 and p. In the original photon direction

$$\text{Initial momentum} = \text{final momentum}$$

$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c}\cos\phi + p\cos\theta \tag{2.16}$$

and perpendicular to this direction

$$\text{Initial momentum} = \text{final momentum}$$

$$0 = \frac{h\nu'}{c}\sin\phi - p\sin\theta \tag{2.17}$$

The angle ϕ is that between the directions of the initial and scattered photons, and θ is that between the directions of the initial photon and the recoil electron. From Eqs. (2.14), (2.16), and (2.17) we can find a formula that relates the wavelength difference between initial and scattered photons with the angle ϕ between their directions, both of which are readily measurable quantities (unlike the energy and momentum of the recoil electron).

The first step is to multiply Eqs. (2.16) and (2.17) by c and rewrite them as

$$pc\cos\theta = h\nu - h\nu'\cos\phi$$

$$pc\sin\theta = h\nu'\sin\phi$$

By squaring each of these equations and adding the new ones together, the angle θ is eliminated, leaving

$$p^2c^2 = (h\nu)^2 - 2(h\nu)(h\nu') \cos \phi + (h\nu')^2 \tag{2.18}$$

Next we equate the two expressions for the total energy of a particle

$$E = \text{KE} + mc^2 \tag{1.20}$$

$$E = \sqrt{m^2c^4 + p^2c^2} \tag{1.24}$$

from Chap. 1 to give

$$(\text{KE} + mc^2)^2 = m^2c^4 + p^2c^2$$

$$p^2c^2 = \text{KE}^2 + 2mc^2\ \text{KE}$$

Since

$$\text{KE} = h\nu - h\nu'$$

we have

$$p^2c^2 = (h\nu)^2 - 2(h\nu)(h\nu') + (h\nu')^2 + 2mc^2(h\nu - h\nu') \tag{2.19}$$

Substituting this value of p^2c^2 in Eq. (2.18), we finally obtain

$$2mc^2(h\nu - h\nu') = 2(h\nu)(h\nu')(1 - \cos\phi) \tag{2.20}$$

This relationship is simpler when expressed in terms of wavelength λ. Dividing Eq. (2.20) by $2h^2c^2$,

$$\frac{mc}{h}\left(\frac{\nu}{c} - \frac{\nu'}{c}\right) = \frac{\nu}{c}\frac{\nu'}{c}(1 - \cos\phi)$$

and so, since $\nu/c = 1/\lambda$ and $\nu'/c = 1/\lambda'$,

$$\frac{mc}{h}\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{1 - \cos\phi}{\lambda\lambda'}$$

Compton effect

$$\lambda' - \lambda = \frac{h}{mc}(1 - \cos\phi) \tag{2.21}$$

Equation (2.21) was derived by Arthur H. Compton in the early 1920s, and the phenomenon it describes, which he was the first to observe, is known as the **Compton effect.** It constitutes very strong evidence in support of the quantum theory of radiation.

Equation (2.21) gives the change in wavelength expected for a photon that is scattered through the angle ϕ by a particle of rest mass m. This change is independent of the wavelength λ of the incident photon. The quantity

Compton wavelength

$$\lambda_C = \frac{h}{mc} \tag{2.22}$$

is called the **Compton wavelength** of the scattering particle. For an electron $\lambda_C = 2.426 \times 10^{-12}$ m, which is 2.426 pm (1 pm = 1 picometer = 10^{-12} m). In terms of λ_C, Eq. (2.21) becomes

Compton effect

$$\lambda' - \lambda = \lambda_C(1 - \cos\phi) \tag{2.23}$$

Compton effect

$$\lambda' - \lambda = \lambda_C(1 - \cos \phi) \tag{2.23}$$

The Compton wavelength gives the scale of the wavelength change of the incident photon. From Eq. (2.23) we note that the greatest wavelength change possible corresponds to $\phi = 180°$, when the wavelength change will be twice the Compton wavelength λ_C. Because $\lambda_C = 2.426$ pm for an electron, and even less for other particles owing to their larger rest masses, the maximum wavelength change in the Compton effect is 4.852 pm. Changes of this magnitude or less are readily observable only in x-rays: the shift in wavelength for visible light is less than 0.01 percent of the initial wavelength, whereas for x-rays of $\lambda = 0.1$ nm it is several percent. The Compton effect is the chief means by which x-rays lose energy when they pass through matter.

de Broglie's Hypothesis

From the study of black body radiation, photo electric effect and Compton effect it was necessary to assign particle like properties to radiation. The amount of energy associated with the radiant particle or photon is

$$E = h\upsilon = \frac{hc}{\lambda} \qquad (1)$$

where h, c, υ and λ are Planck's constant, speed of light, frequency and wavelength, respectively, and since $\upsilon\lambda = c$. Eqn. (1) proves that the radiation has particle like nature described by photons and, at the same time, wave like nature described by wavelength λ.

Louis de Broglie asserted that this dual property is not merely confined to photon or radiation, it can be attributed to any material particle. For his hypothesis de Broglie relied on the special theory of relativity and old quantum theory. These waves that arise out of matter particles are known as matter waves.

The energy E and momentum p of the photon is related by

$$p = \frac{E}{c} = \frac{h\upsilon}{c} = \frac{h}{\lambda}$$

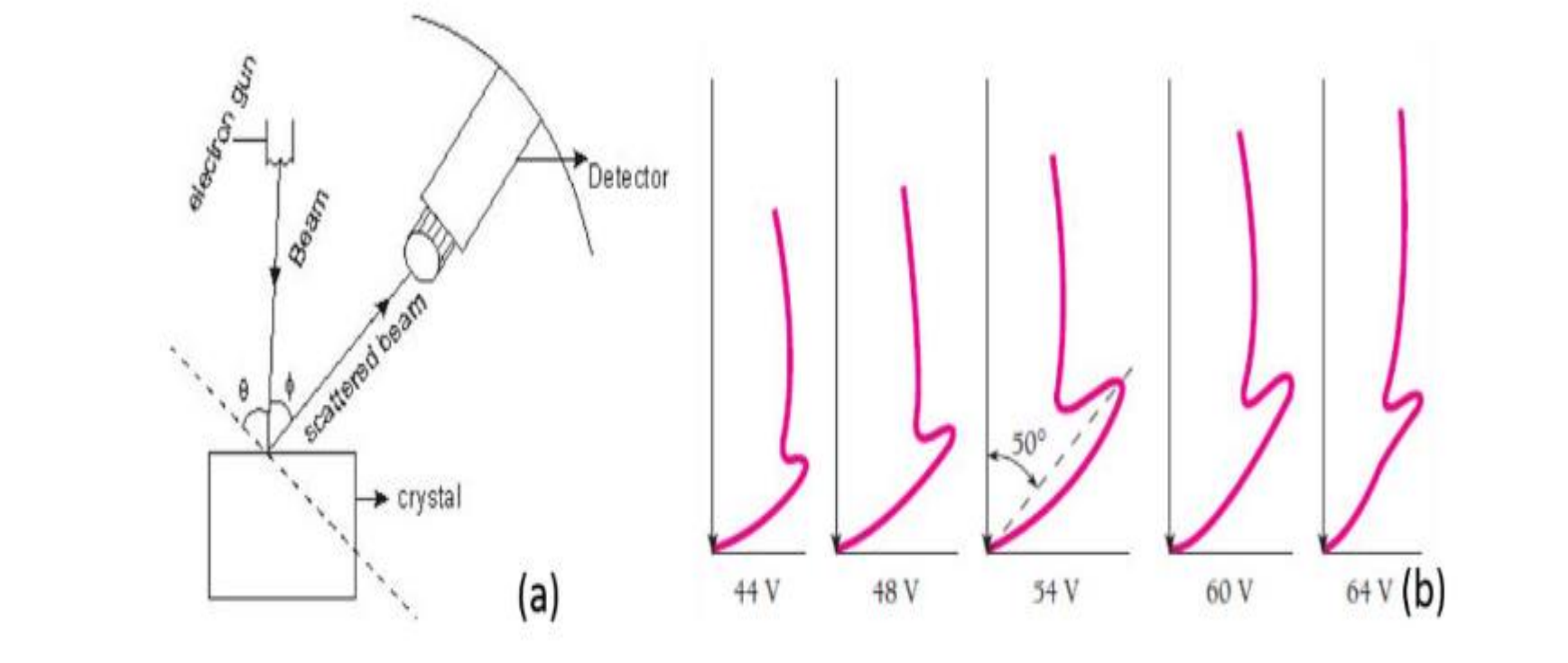
The wavelength of a photon is therefore specified by its momentum according to the relation

$$\lambda = \frac{h}{p} \qquad (2)$$

de Broglie asserted that the nature is symmetric. Therefore, what is applicable to photon can also be applied to any other particle.

1. Davisson-Germer Experiment (*Experimental verification of de Broglie hypothesis*)

Clinton Davisson and Lester Germer in the USA and G.P. Thompson in the UK independently confirmed de Broglie's hypothesis experimentally. Davisson and Germer studied the diffraction



The experimental arrangement is shown in Fig. 1a. The electrons from a hot tungsten electron gun are collimated and incident normally on the crystal surface. The intensities of the electron beam scattered in all directions by the atoms of the crystal, which are arranged in a regular periodic fashion, were measured as a function of the angle of scattering. The results are shown in polar diagram (Fig.1b). The length of the radius vector is proportional to the intensity of the scattered beam and the angle of the radius vector and the Y-axis is the angle of scattering. Experimental results show that the length of the peak increases until it reaches a maximum at $\phi = 50°$ for $V = 54$ V. This can be explained as follows.

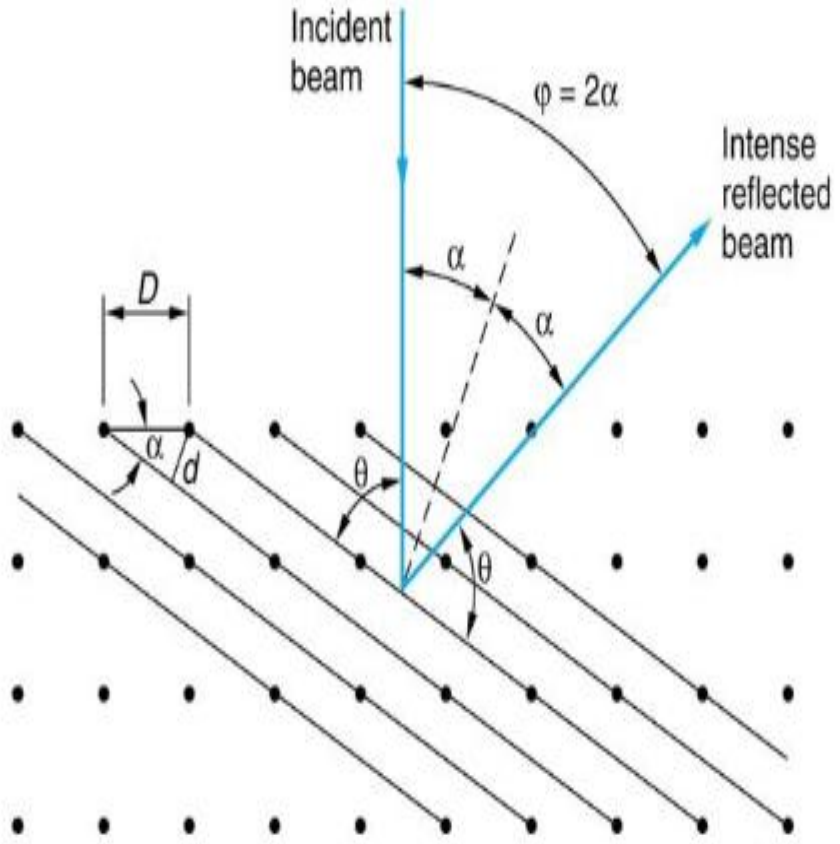


Fig. Schematic representations of atomic planes for Davison and Germer experiment.

of electron beam using nickel crystal. If the electrons are accelerated by a potential difference of V volts it gains a kinetic energy given by

$$\frac{1}{2}mv^2 = eV \qquad (4)$$

Where v is the velocity attained by the electrons and $v << c$, c being the speed of light. The de Broglie wavelength of the electron is

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{(2meV)}} \qquad (5)$$

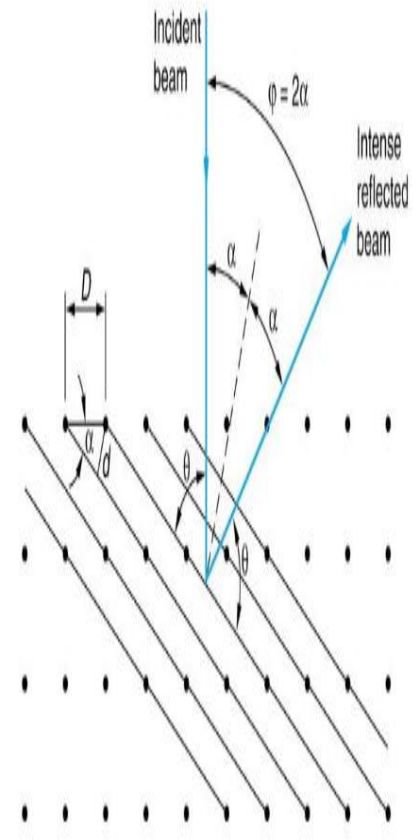


Fig. Schematic representations of atomic planes for Davison and Germer experiment.

The planes are shown in Fig. 2, where D is the distance between the consecutive atoms and d is the inter-planar separation. The Bragg's condition for constructive interference is

$$n\lambda = 2d\sin\theta = 2d\sin(90^\circ-\alpha) = 2d\cos\alpha \qquad (6)$$

Again, from the above figure,

$$d = D\sin\alpha \qquad (7)$$

Combining Eqns. 6 and 7, we get

$$n\lambda = 2D\sin\alpha.\cos\alpha = D\sin 2\alpha$$

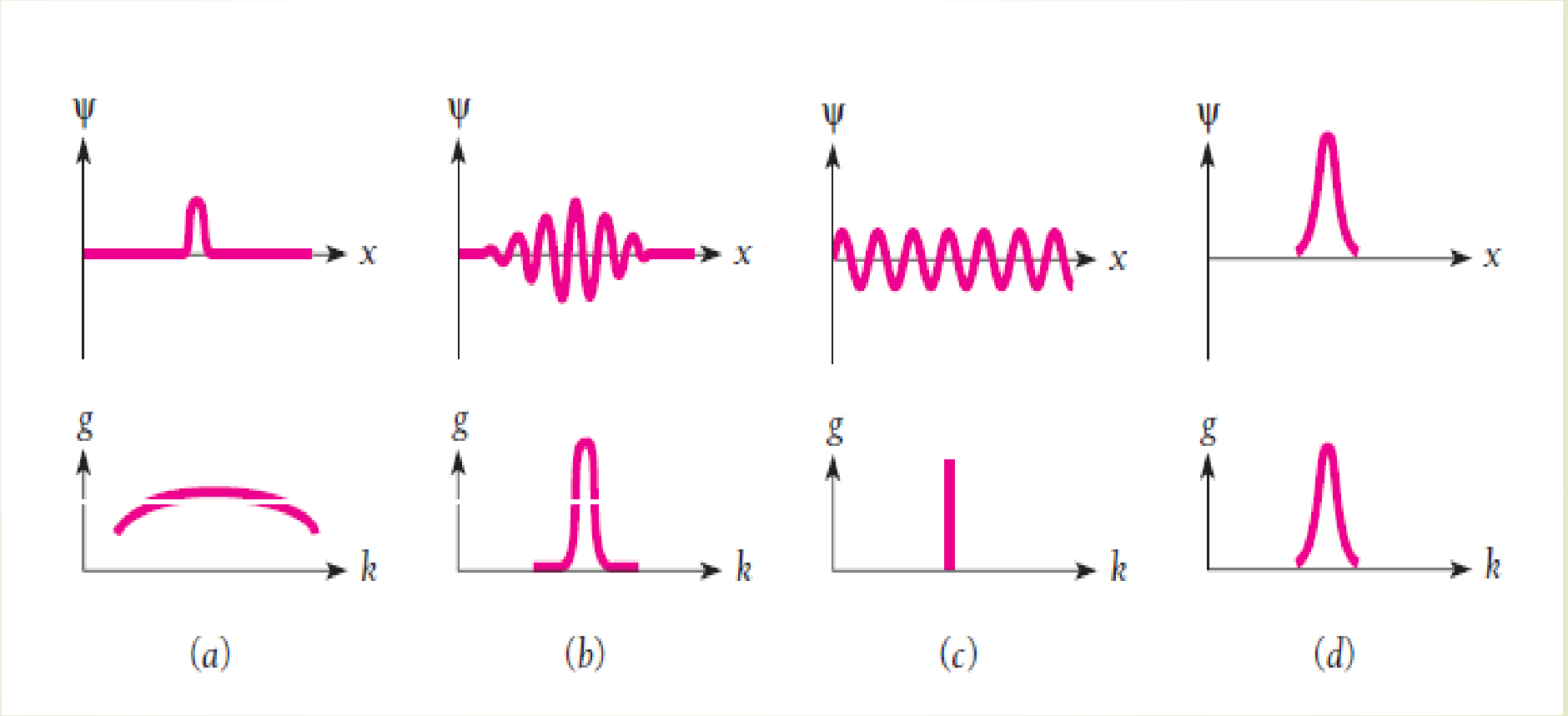
In terms of ϕ

$$n\lambda = D\sin\phi \qquad (8)$$

From x-ray data it is known that for nickel D = 2.15 Å. Therefore, for 54 V electron at ϕ= 50° and in the first order (i.e. n = 1)

$$\lambda = 2.15\ \text{Å}.\ \text{Sin}\ 50^\circ = 1.65\ \text{Å}$$

Further, with V =54 V and standard values of the constants Eqn. 5 can be used to obtain the wavelength of electron as λ = 1.66 Å. The remarkably close agreement between the two results proves the validity of de Broglie hypothesis.


ψ
x
g
k
(a)
(b)
(c)
(d)

HEISENBERG'S UNCERTAINTY PRINCIPLE

A moving particle is associated with de Broglie wave group. So it is better to represent it by a group of waves rather than by a localized entity. So there is a fundamental limit to the accuracy with which we can measure the properties of the particle, such as its momentum and position. When the de Broglie wave group is very narrow the position of the particle can be easily estimated within the width of the group, but the wavelength is very hard to establish (Fig. 1). If the wide of the group is wide, then wavelength can be easily established, but the particle may be anywhere within the wide width of the group (Fig. 2).

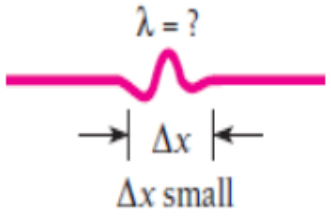


Fig. 1

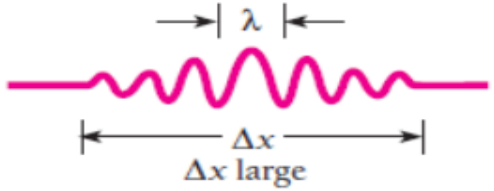


Fig. 2

Thus there is always an uncertainty in the simultaneous measurement of the momentum and position of the particle. The uncertainty principle states that 'The product of the uncertainty δx in the position of the body at some instant and uncertainty δp in its momentum at the same instant is equal to or greater than $h/2\pi$, i.e. the reduced Planck's constant ($\hbar$), h being Planck's constant'. i.e.

$$\delta x \, \delta p \geq h/2\pi$$

Applications of Heisenberg's uncertainty principle

1 Non-Existence of Electron in the Nucleus

The radius of the nucleus of an atom is of the order of 10^{-14} m. If an electron is confined within the nucleus, the uncertainty in its position must not be greater than the diameter, i.e. 2 x 10^{-14} m, of the nucleus. According to uncertainty principle for the lowest limit of accuracy

$$\Delta x \, \Delta P = \frac{h}{2\pi}$$

Where Δx *is* uncertainty in the position and ΔP is the uncertainty in the momentum.

$$\Delta P = \frac{h}{2\pi \Delta X} = \frac{6.625 \times 10^{-34}\,[J-s]}{2 \times 3.14 \times 2 \times 10^{-14}[m]} \quad (\text{as } \Delta X = \text{diameter of nucleus})$$

$$\Delta P = 5.275 \text{ x } 10^{-21} \text{ kg m/sec}$$

This is the uncertainty in momentum of the electron. It means the momentum of the electron would not be less than ΔP, rather it could be comparable to ΔP. Thus

$$\Delta P = 5.275 \text{ x } 10^{-21} \text{ kg m/sec}$$

The kinetic energy of the electron can be obtained in terms of momentum as

$$T = \tfrac{1}{2} mv^2 = p^2 / 2m$$

$$= \frac{(5.275 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31}} J$$

$$= \frac{(5.275 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} eV$$

$$= 95.55 \times 10^6 eV$$

$$\approx 96MeV$$

From the above result, it is clear that the electrons inside the nucleus may exist only when it possesses the energy of the order of 96 MeV. However, the maximum possible kinetic energy of an electron emitted by radioactive nuclei has been found about 4MeV. Hence, it is concluded that the electron cannot reside in the nucleus.